Tathagata Karmakar

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SUMMARY

Quantum information researcher specializing in quantum optimal control, quantum algorithms, resource estimation, and high-performance quantum simulation.

TECHNICAL EXPERTISE

- **AI/ML:** PyTorch, JAX, neural operators, physics-informed ML.
- **Quantum Algorithms and Applications:** Measurement-driven algorithms, SAT/optimization, multi-qubit open quantum systems simulation (unitary + dissipative).
- **Optimal control:** Pontryagin, gradient-based, hardware-constrained control optimization.
- **HPC and Software:** Mathematica, Python, Numba, MPI, QuTiP, mpi4py.

RESEARCH EXPERIENCE

Postdoctoral Scholar — K. B. Whaley group – UC Berkeley (2024–Present)

Quantum Control, Algorithms & Simulation

- Developed measurement-driven quantum k-SAT solvers and designed scalable numerical pipelines capable of simulating 20-qubit dynamics.
- Designed real-time optimal control protocols for measurement-driven computation, improving query complexity.
- Created noise-canceling feedback strategies improving success rates in state preparation.
- Designed feedback protocols for **5-to-1 magic state distillation**, achieving **300–400% improvement** in successful distillation probability.

Graduate Researcher — University of Rochester (2018–2024)

Quantum Control, Open Quantum Systems, and Stochastic Methods

- Designed real-time **optimal control protocols** for oscillator-encoded qubits (binomial codes, cat-state transformations, parametric cooling).
- Engineered cost functions and control schedules, achieving 40–190% performance improvements across multiple state-preparation tasks.
- Built numerical pipelines (JAX/Numba) for quantum optimal control protocols, benchmarked using fidelity metrics with trajectory simulations.
- Collaborated on the experimental realization of superresolving imaging systems.

Research Intern — PHI Lab, NTT Research, Inc. (2023)

Machine-Learning-Based Model Reduction for Optical Systems

- Built a **neural operator** architecture for nonlinear optical systems, learned dynamics across **256 initial conditions**, implemented **GPU-accelerated** training.

SELECTED PUBLICATIONS

- **Optimal schedule for solving the k-SAT**, arXiv:2507.16128 (2025).
- **Noise-Canceling Quantum Feedback**, Phys. Rev. A **113**, 042403 (2026).
- **CDJ-Pontryagin Optimal Control**, Quantum **10**, 2043 (2026).
- **Stochastic path-integral analysis**, PRX Quantum **3**, 010327 (2022).

  Additional publications in **Phys. Rev. Research, Optics Express, J. Phys. A**.

EDUCATION

Ph.D., Physics & Astronomy — University of Rochester (2024).

B.S., Physics — IIT Kanpur (2018), CPI: 9.9/10.